

Rule Catalog

Design Reminder

The experiment consists of six sessions with five blocks each, one per rule family: Arithmetic, Attraction, Recoloring, Expansion, and Occlusion.

Each block contains one rule family and one of two contexts:

- **Previous (Inference):** compare the current rule with the rule from the previous trial.
- **First (Application):** compare the current rule with the rule introduced at the start of the block.

Each trial presents two examples of the same hidden rule. A block contains one initial rule trial followed by eight same/different decisions, balanced to four *same* and four *different* trials with a maximum run length of three identical responses.

Contexts alternate within sessions and the starting context alternates across sessions. Rule families are assigned such that each family occurs equally often in both contexts across the six sessions.

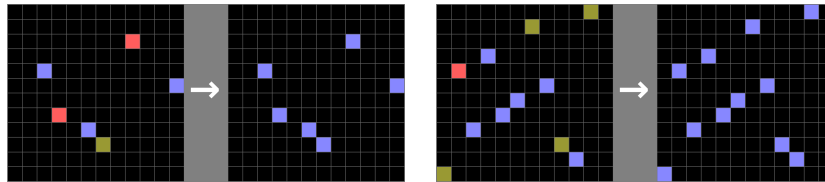
Frame colour and button mapping are counterbalanced across four participant versions (see experimenter protocol for details).

Trials last 10 s and each block is preceded by 10 s fixation. Including the final fixation and 10 s of dummy scans (e.g. $TR = 2$ s), each session requires 520 s (8 min 40 s; 260 volumes).

Arithmetic

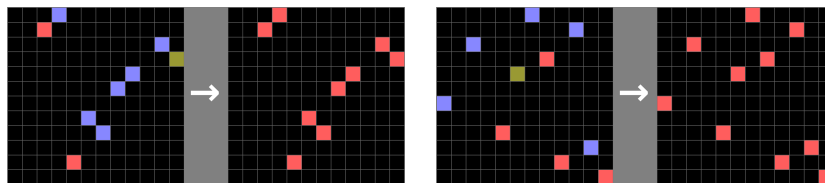
- Majority takeover

Event: recoloring Condition: color, counting Stimulus: dots



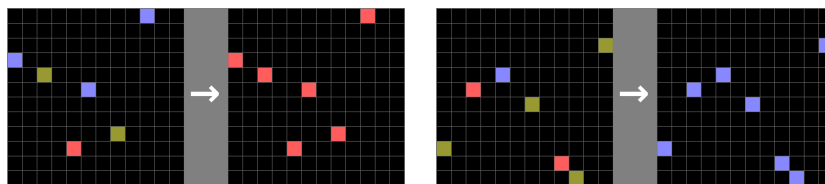
- Middle takeover

Event: recoloring Condition: color, counting Stimulus: dots



- Minority takeover

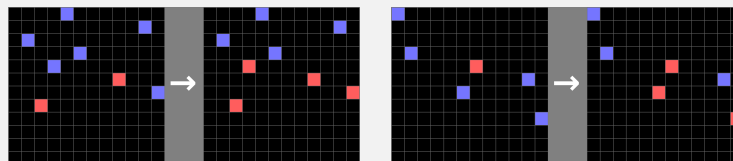
Event: recoloring Condition: color, counting Stimulus: dots



- Equalize colors

Excluded

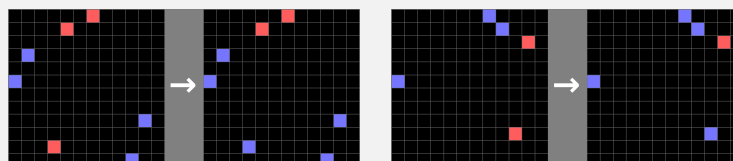
Event: recoloring Condition: color, counting Stimulus: dots



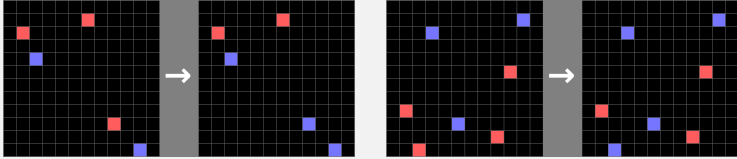
- Increment majority color

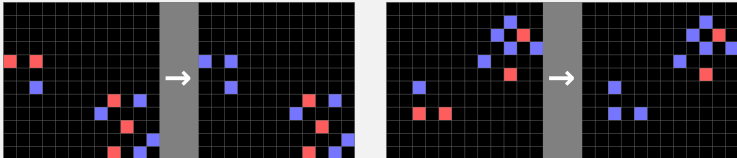
Excluded

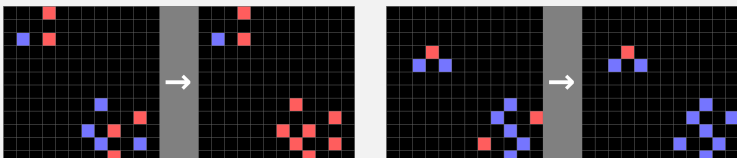
Event: recoloring Condition: color, counting Stimulus: dots

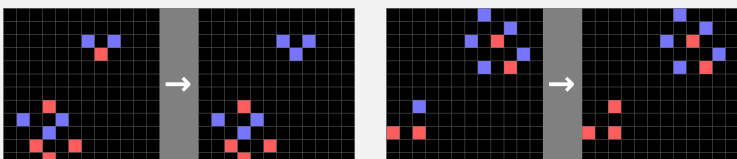


- Increment minority color** *Excluded*
Event: recoloring **Condition:** color, counting **Stimulus:** dots


- Majority in big overtakes** *Excluded*
Event: recoloring **Condition:** color, counting **Stimulus:** clustered_dots

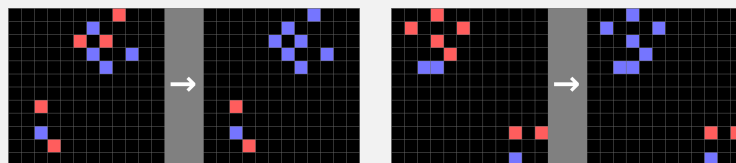

- Majority in small overtakes** *Excluded*
Event: recoloring **Condition:** color, counting **Stimulus:** clustered_dots


- Minority in big overtakes** *Excluded*
Event: recoloring **Condition:** color, counting **Stimulus:** clustered_dots



- Minority in small overtakes
 Excluded

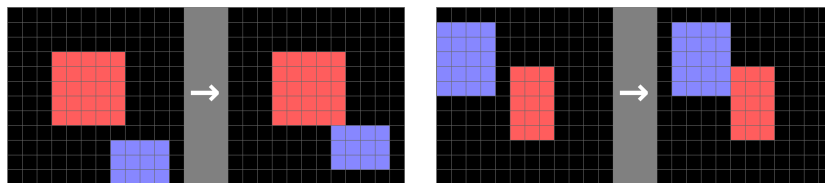
 Event: recoloring Condition: color, counting Stimulus: clustered_dots



Attraction

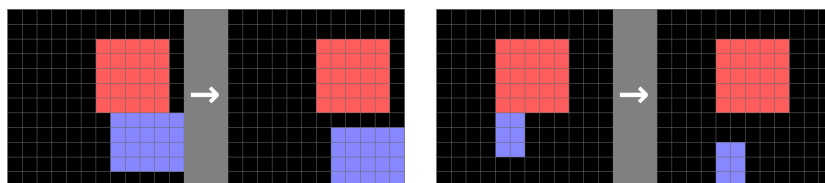
- Color attraction

Event: attraction Condition: color, movement Stimulus: big_blocks



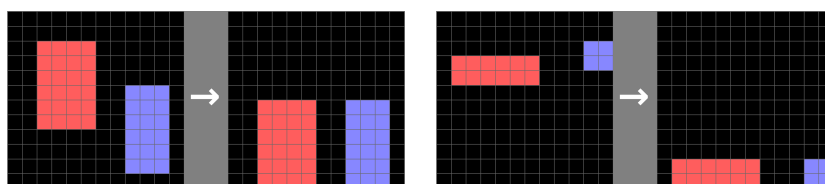
- Color repulsion

Event: repulsion Condition: color, movement Stimulus: big_blocks



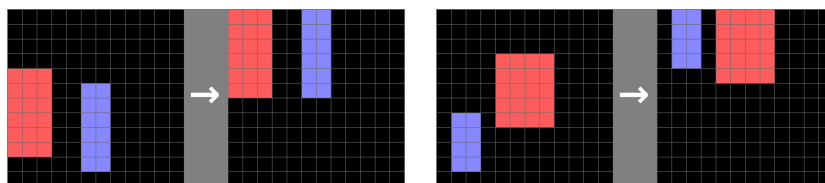
- Falling blocks

Event: falling Condition: movement Stimulus: big_blocks



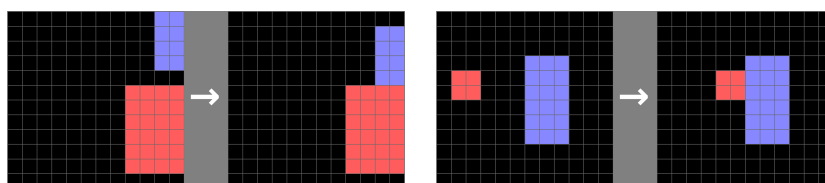
- Floating blocks

Event: floating Condition: movement Stimulus: big_blocks



- Size attraction

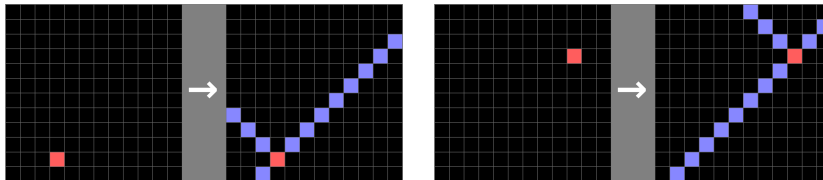
Event: attraction Condition: shape, movement Stimulus: big_blocks



Expansion

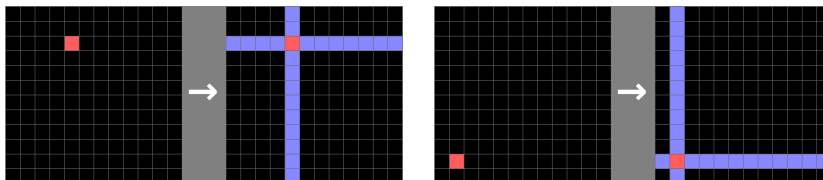
- 3 arm star ray

Event: expansion Condition: shape Stimulus: ray_expansion



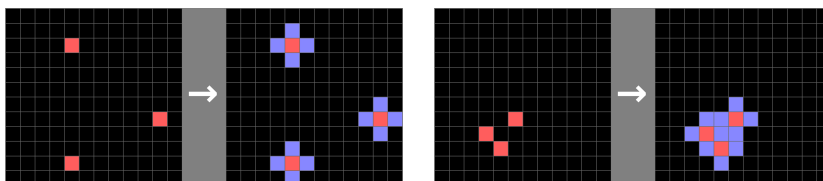
- Plus ray

Event: expansion Condition: shape Stimulus: ray_expansion



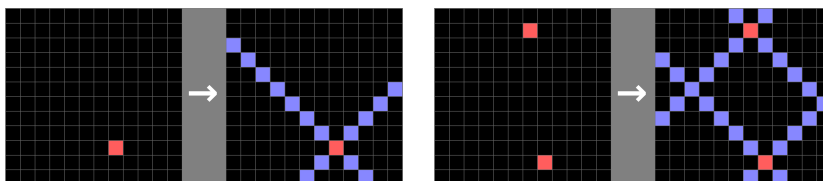
- Plus step

Event: expansion Condition: shape Stimulus: step_expansion



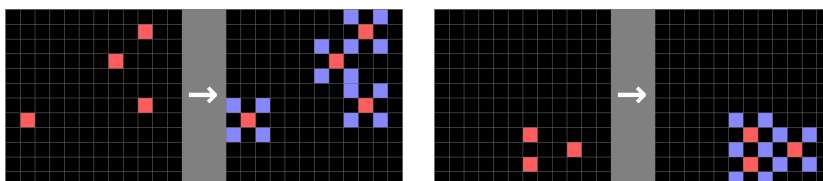
- Star ray

Event: expansion Condition: shape Stimulus: ray_expansion



- Star step

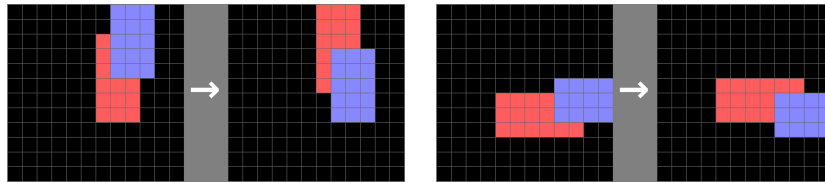
Event: expansion Condition: shape Stimulus: step_expansion



Occlusion

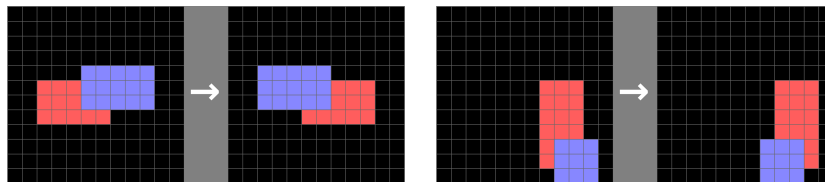
- Mirror x

Event: mirror Condition: shape Stimulus: occluding_blocks



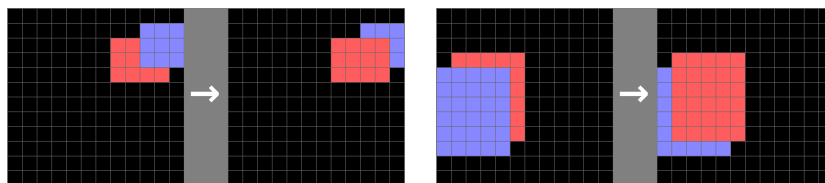
- Mirror y

Event: mirror Condition: shape Stimulus: occluding_blocks



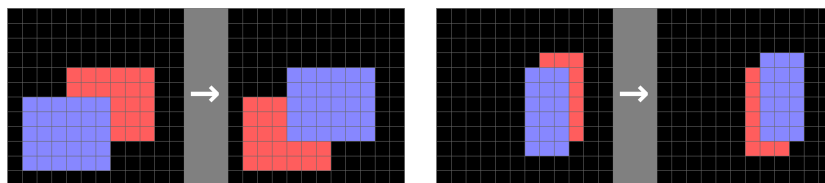
- Occlusion reversal

Event: occlusion_reversal Condition: shape Stimulus: occluding_blocks



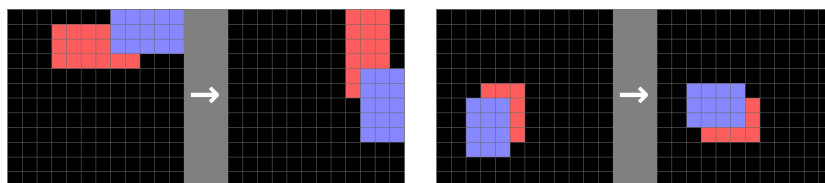
- Rotate 180

Event: rotation Condition: shape Stimulus: occluding_blocks



- Rotate 90

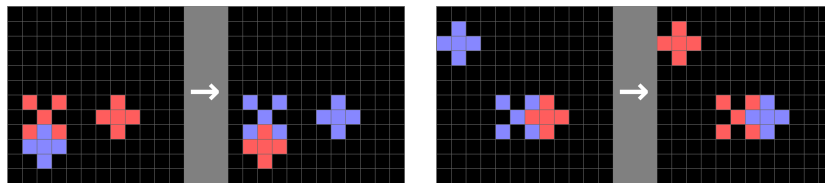
Event: rotation Condition: shape Stimulus: occluding_blocks



Recolor

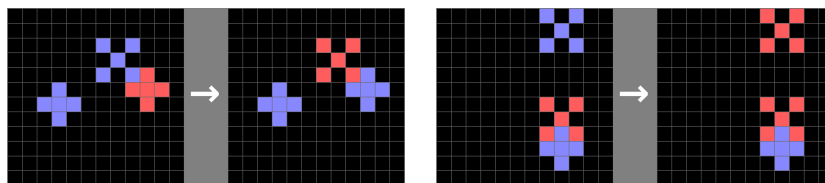
- Color inversion

Event: recoloring Condition: color Stimulus: cross_plus



- Shape color mapping

Event: recoloring Condition: shape Stimulus: cross_plus



- Touching edges recolor

Event: recoloring Condition: shape, neighbor Stimulus: cross_plus

